D. Y. Patil Agricultural and Technical University

Talsande

PROJECT REPORT ON

Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization

Department

Master Of Computer Applications

(MCA-Sandwich Sem-II)

Submitted By

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INTRODUCTION

1.1 Project Overview

Businesses use data analysis to enhance their marketing and sales strategies in the cutthroat business world of today. Product placement has a significant impact on what customers decide to buy. Placing products in a way that draws customers and boosts sales is known as strategic product placement.

In order to comprehend how product placement affects sales performance, this project focuses on analyzing sales data. Raw sales data can be converted into interactive dashboards and insightful information by utilizing data visualization programs like Tableau. These dashboards make it simple for companies to spot patterns, trends, and consumer preferences.

This project analyzes sales data according to time, location, and product category. Businesses can better understand which products perform well and how product placement tactics can increase total revenue by using the visualization.

1.2 Purpose

This project's goal is to analyze sales data and ascertain how product placement tactics impact sales results. Tableau is used in the project to build interactive dashboards that show the connections between locations, products, and consumer purchasing patterns.
The primary objective is to offer information that enables companies to make better choices about product placement, marketing tactics, and sales scheduling. Organizations can optimize their product positioning and enhance overall business performance by knowing which products generate higher sales and where they perform best.

IDEATION PHASE

2.1 Problem Statement

Many companies gather a lot of sales data, but they find it difficult to draw meaningful conclusions from it. Businesses are unable to determine which products are performing well or how product placement affects sales without conducting adequate analysis.

As a result, rather than using data-driven insights, businesses might base their decisions on conjecture. Missed business opportunities, poor sales performance, and ineffective product positioning could result from this.

By analyzing sales data and utilizing Tableau dashboards to present insights, this project seeks to address this issue. These graphics aid in determining sales trends and assessing how well product placement tactics are working.

2.2 Empathy Map Canvas

Understanding the requirements and expectations of stakeholders, including analysts, marketing teams, and business managers, is made easier with the use of the empathy map.

Think:
Managers want to improve product visibility and increase sales.

Feel:
They feel uncertain about which product placement strategies work best.

Say:
They often say that better insights from data can help them make better business decisions.

Do:
They analyse sales reports, monitor product performance, and adjust marketing strategies.

Pain:
Large datasets are difficult to analyse manually, and valuable insights may be missed.

Gain:
Interactive dashboards and visual analytics help them quickly understand sales patterns and optimize product placement strategies.

2.3 Brainstorming

A number of concepts to enhance the comprehension of product placement and sales performance were produced during brainstorming. Identifying best-selling products, analyzing sales trends by product category, researching regional sales performance, and visualizing monthly or annual sales growth are some of the key concepts. To determine which products are the most profitable, another strategy is to analyze profit margins and compare them with sales performance.
Tableau dashboards facilitate the visual presentation of these insights, allowing decision-makers to swiftly analyze the data and take appropriate action.

REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The customer journey map represents the steps involved in analysing sales data and generating insights.

1. The process begins with collecting sales data from available datasets.
2. The collected data is cleaned and prepared for analysis by removing errors and missing values.
3. The processed dataset is then imported into Tableau.
4. Various visualizations such as bar charts, line charts, and dashboards are created.
5. Users interact with the dashboard to explore sales trends and patterns.
6. Based on these insights, businesses can improve product placement strategies.

3.2 Solution Requirement

Functional Requirements

- Import and manage sales datasets.
- Create interactive dashboards using Tableau.
- Visualize sales performance across product categories.
- Analyse regional and time-based sales trends.
- Provide filters and interactive features for deeper analysis.

Non-Functional Requirements

- The system should be easy to use and understand.
- Dashboards should display data quickly and accurately.
- Visualizations should be clear and informative.
- The system should support large datasets efficiently.

Data Flow Diagram

Level 0 DFD

User → Dataset → Tableau Visualization → Sales Insights → Decision Making

Level 1 DFD

Dataset → Data Cleaning → Data Processing → Tableau Dashboard → Business Analysis

Technology Stack

Technology	Description
Tableau 2026.1	Used for data visualization and dashboard creation
Excel	Used as the dataset source
Python (Flask)	Used for creating web app
GitHub	Used for storing project files and version control
Web Browser	Used to access dashboards and project repository

PROJECT DESIGN

4.1 Problem Solution Fit

By transforming unprocessed sales data into visual insights, the project tackles the challenge of comprehending the efficacy of product placement. Tableau dashboards make it simple for users to analyze sales patterns and determine which products bring in the most money. This aids companies in making well-informed choices regarding their product placement tactics.

4.2 Proposed Solution

Making an interactive Tableau data visualization dashboard is the suggested remedy. Numerous sales metrics, including product category performance, regional sales distribution, profit analysis, and sales trends over time, will be shown on the dashboard.

Users can analyze trends and spot chances to enhance product placement tactics with the aid of these visualizations.

4.3 Solution Architecture

Data Source (Sales Dataset)

↓

Data Cleaning and Preparation

↓

Tableau Visualization and Dashboard Creation

↓

Sales Analysis and Insights

↓

Business Decision Making

PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was divided into several phases to ensure proper implementation.

1. Data collection from online datasets.
2. Data cleaning and preprocessing.
3. Importing dataset into Tableau.
4. Creating charts and dashboards.
5. Testing and validating visualizations.
6. Publishing the project on GitHub and Tableau Public.

FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

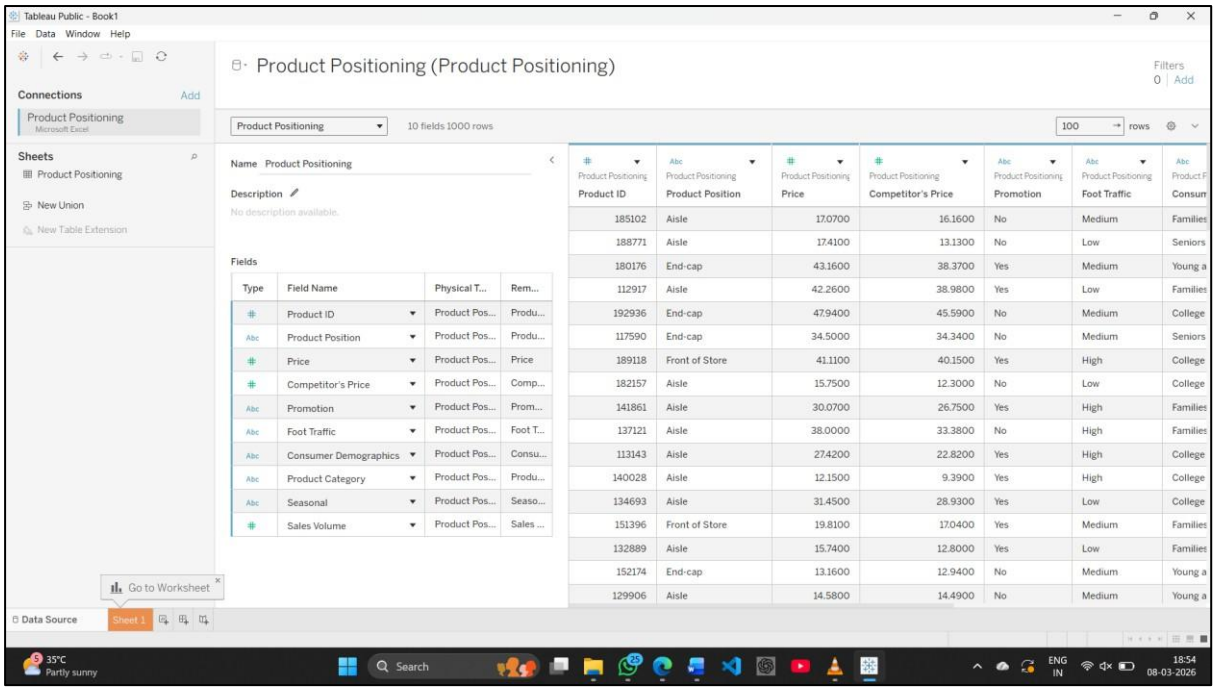
Test Case	Expected Result
Load dataset	Dataset loads successfully
Create visualization	Charts display correct values
Apply filters	Dashboard updates dynamically
Analyse trends	Sales patterns are clearly visible

RESULTS

7.1 Output Screenshots

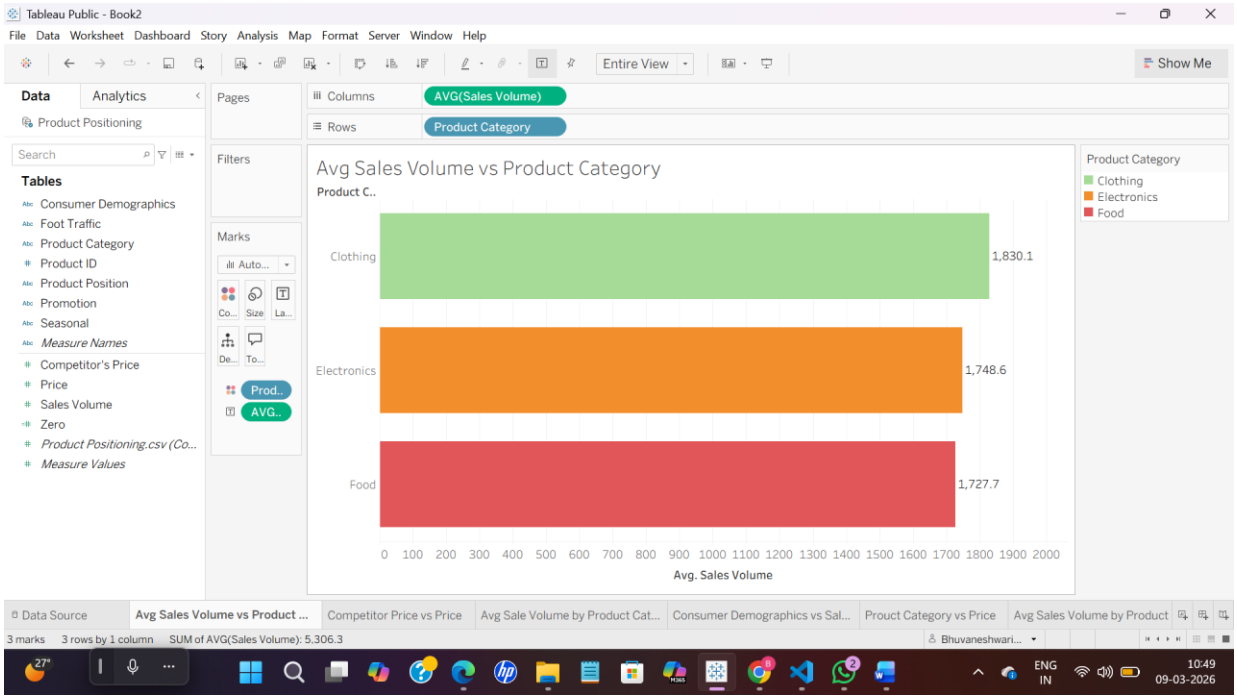
The project creates interactive dashboards that show a variety of insights, including regional sales distribution, product category performance, and sales trends. This section includes screenshots of the Tableau dashboard, charts, and analysis results to show the project's output.

- **Data Collection and Extraction:**

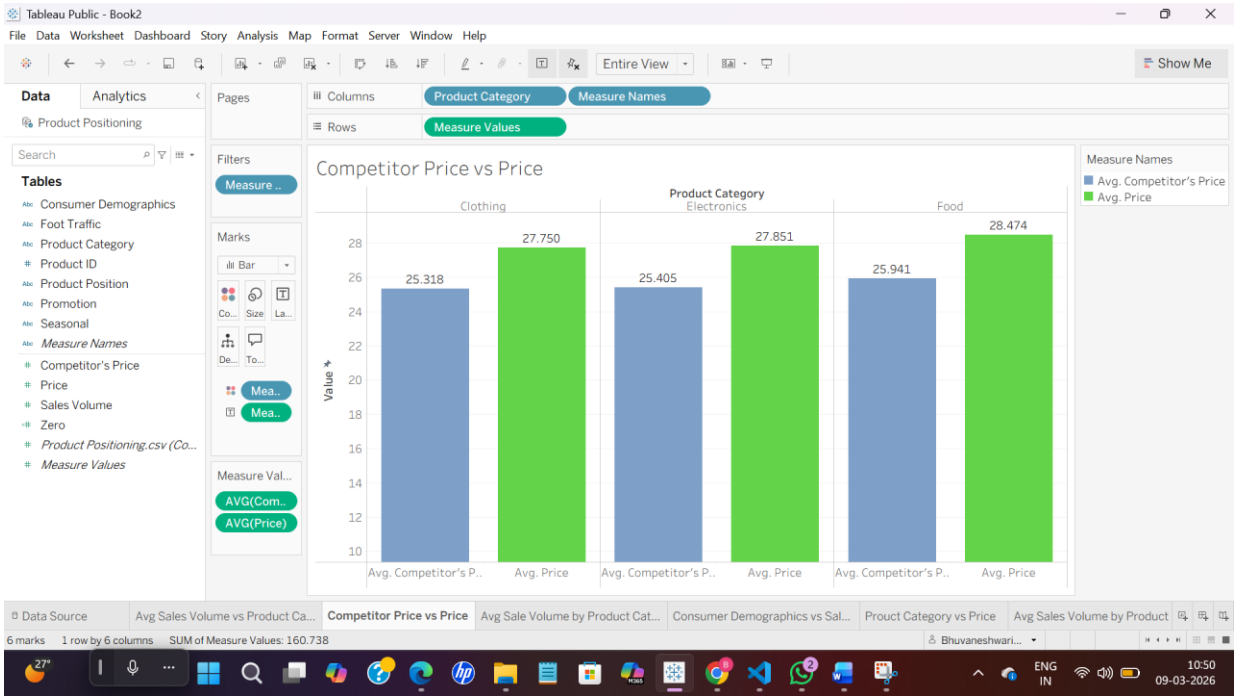


- **Data Preparation and Data Visualization**

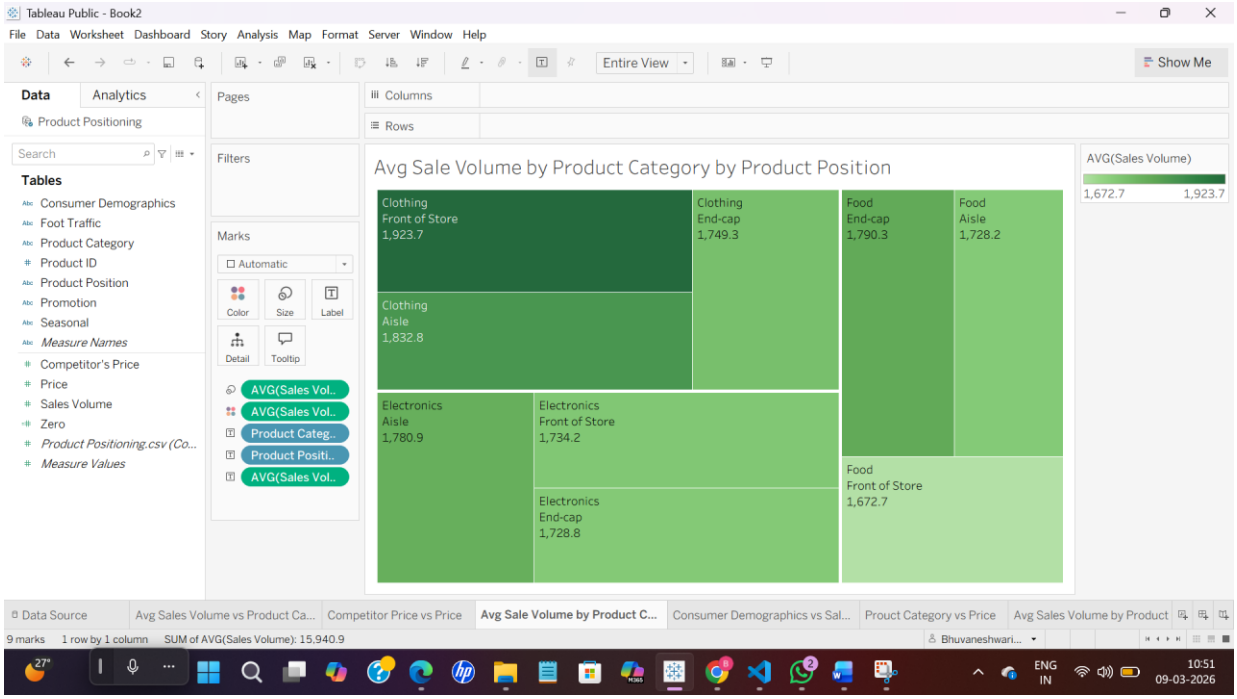
Sheet 1 - Avg Sales Volume vs Product Category



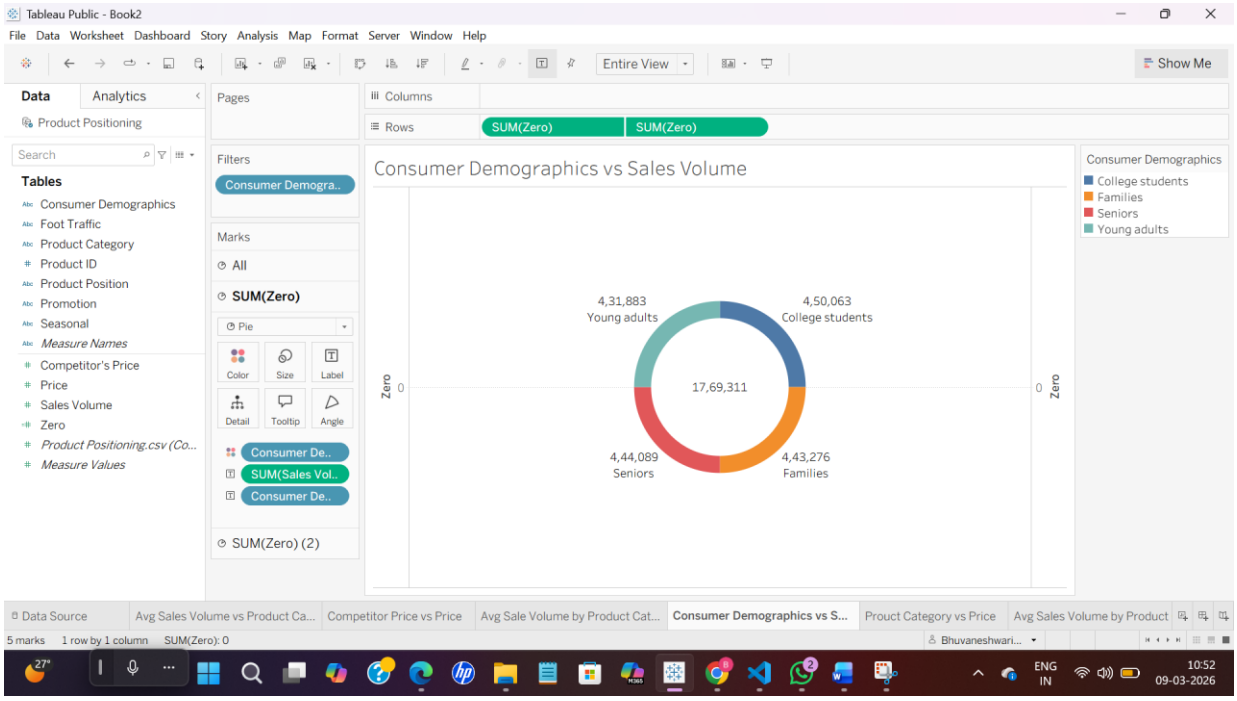
Sheet 2 - Competitor Price vs Price



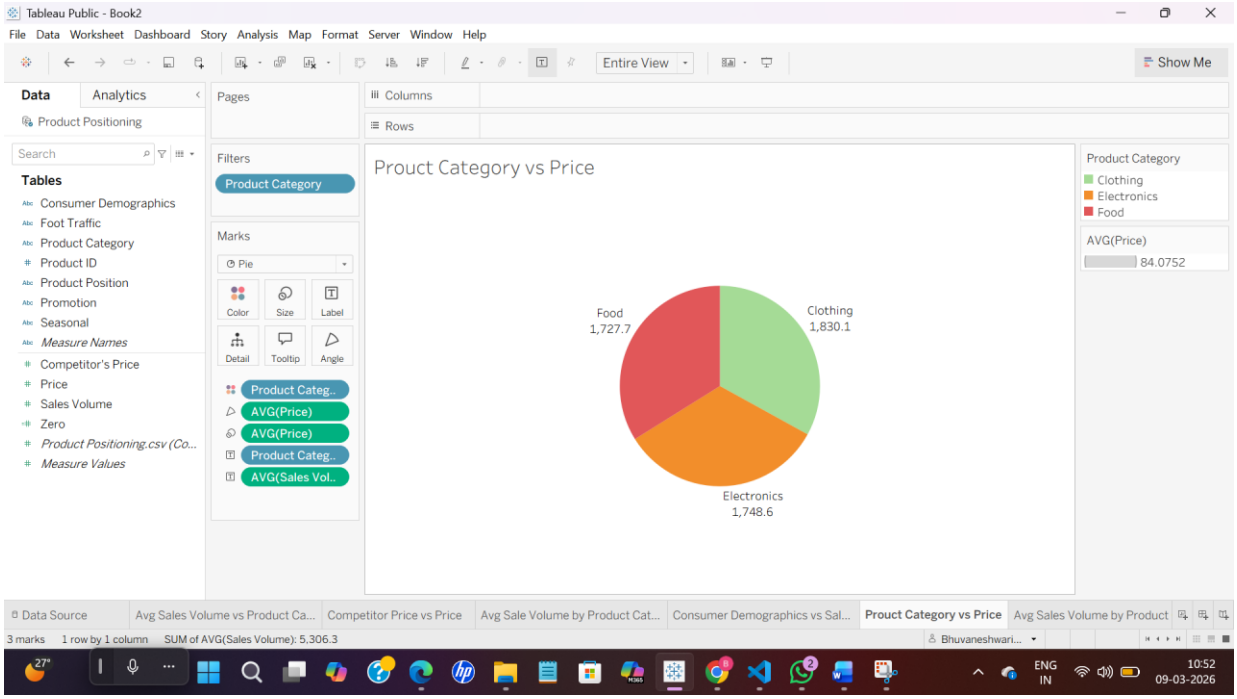
Sheet 3 - Avg Sales Volume by Product Category by Product Position



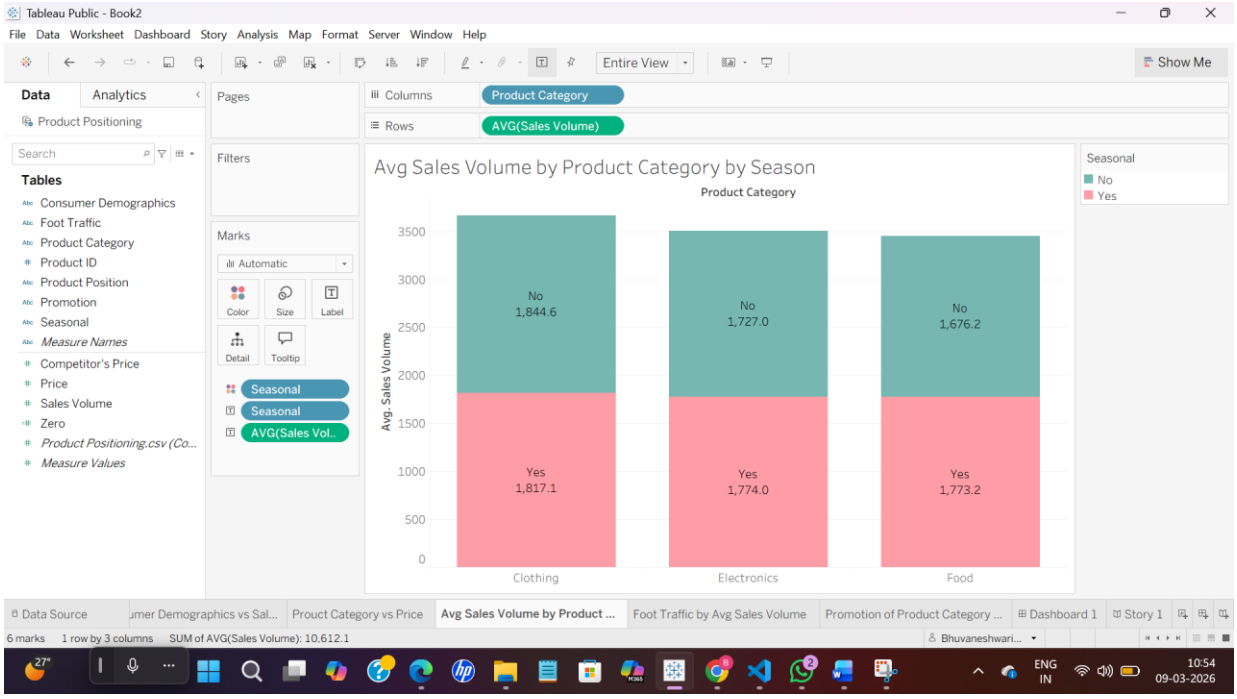
Sheet 4 - Consumer Demographics vs Sales Volume



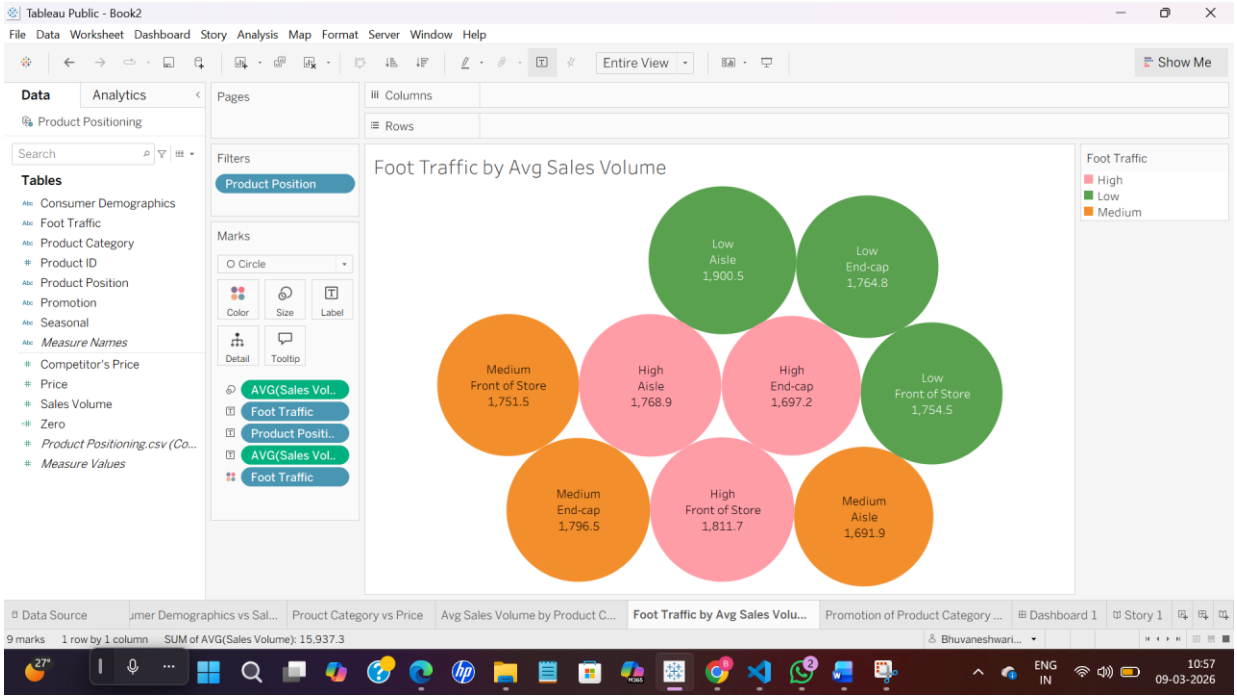
Sheet 5 - Product Category vs Price



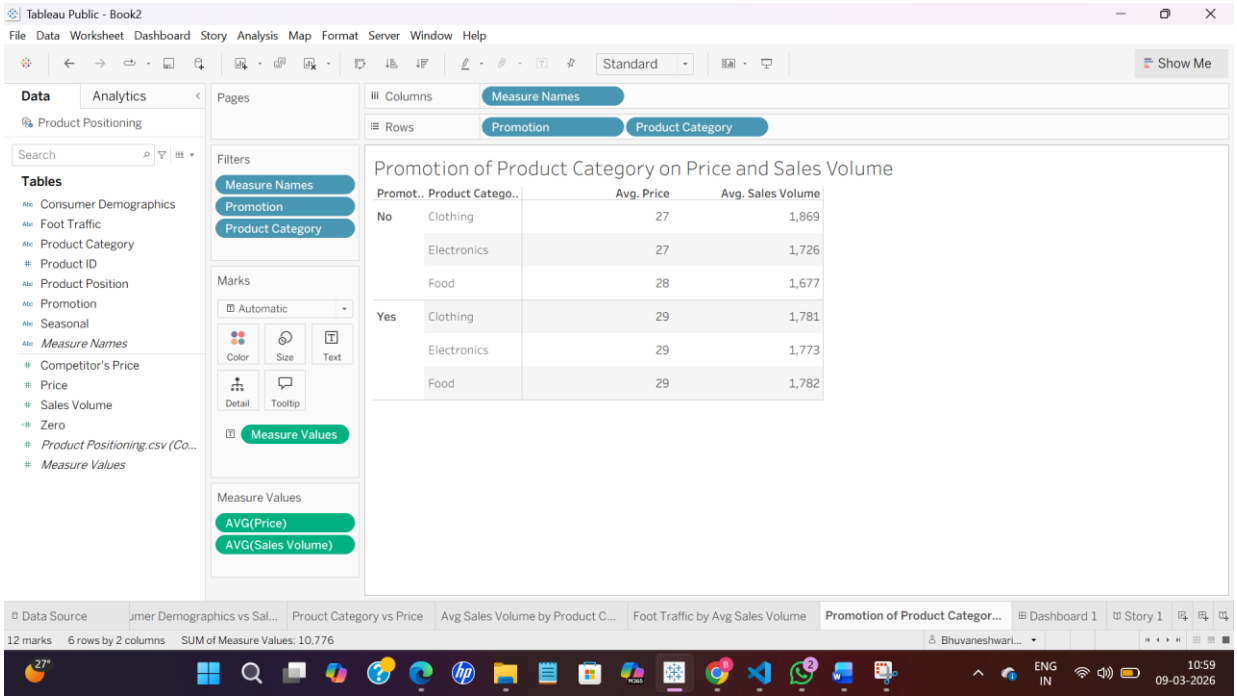
Sheet 6 - Avg Sales Volume by Product Category by Season



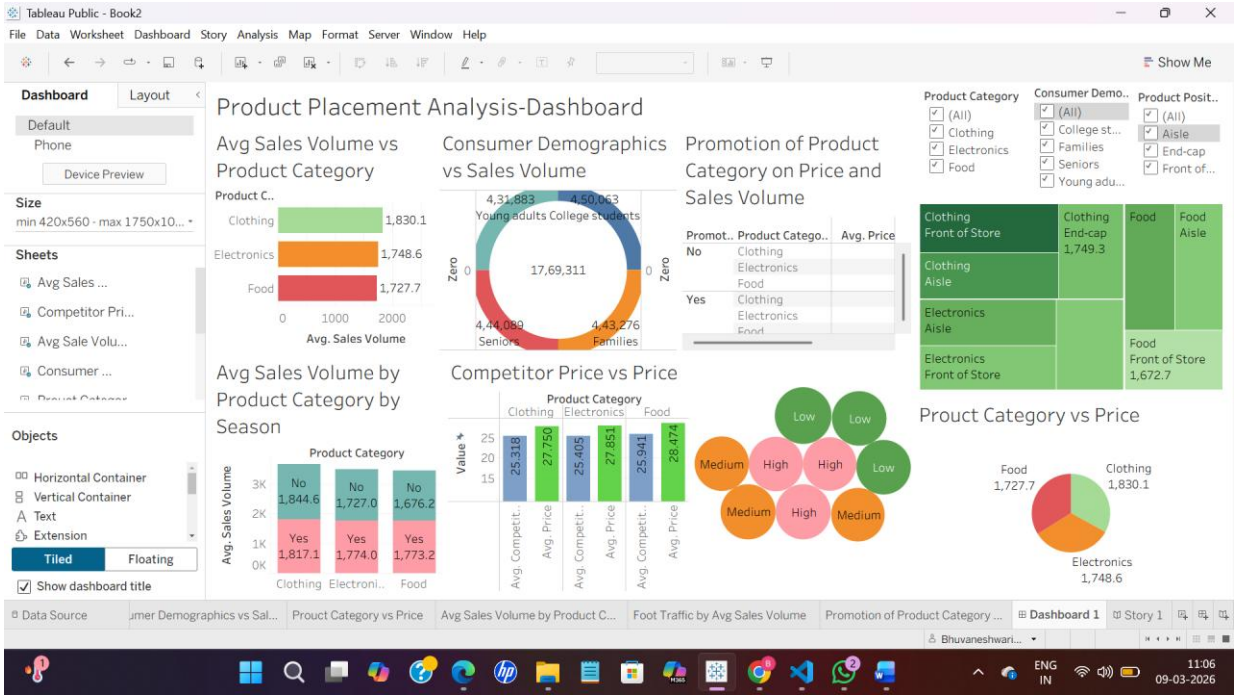
Sheet 7 - Foot Traffic by Avg Sales Volume



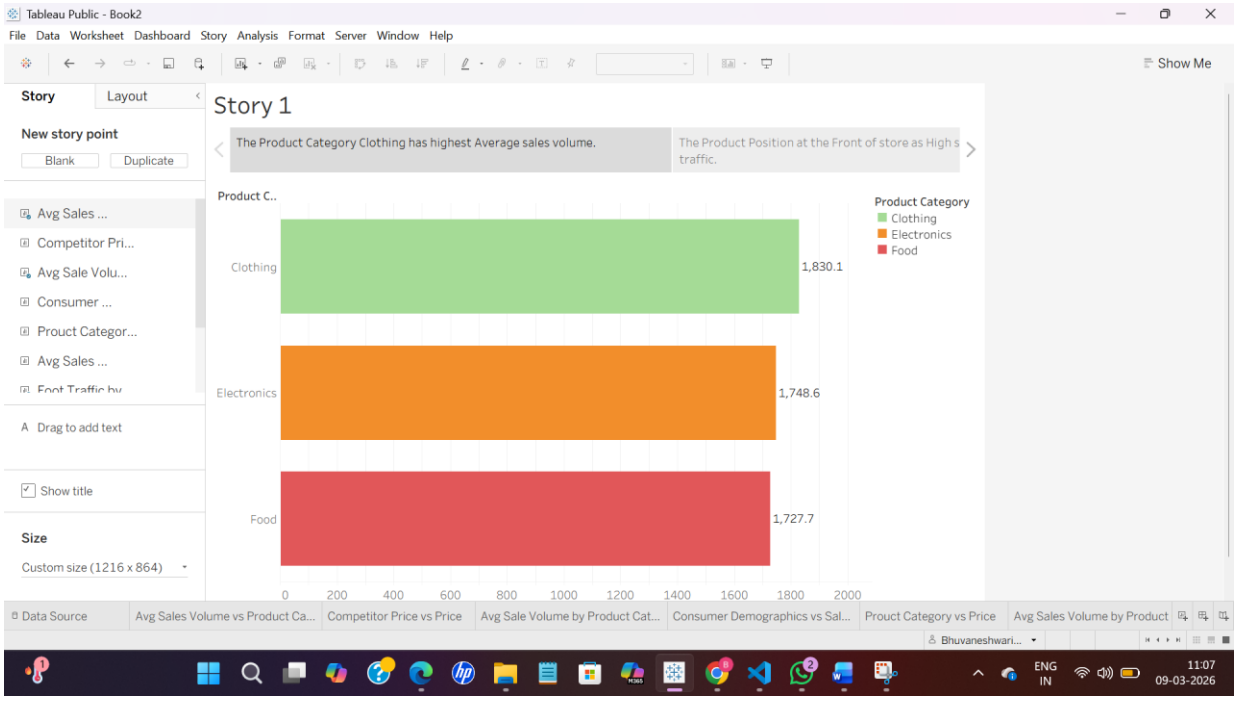
Sheet 8 - Promotion of Product Category on Price and Sales Volume



- **Design of Dashboard**



- **Scenes of Story**



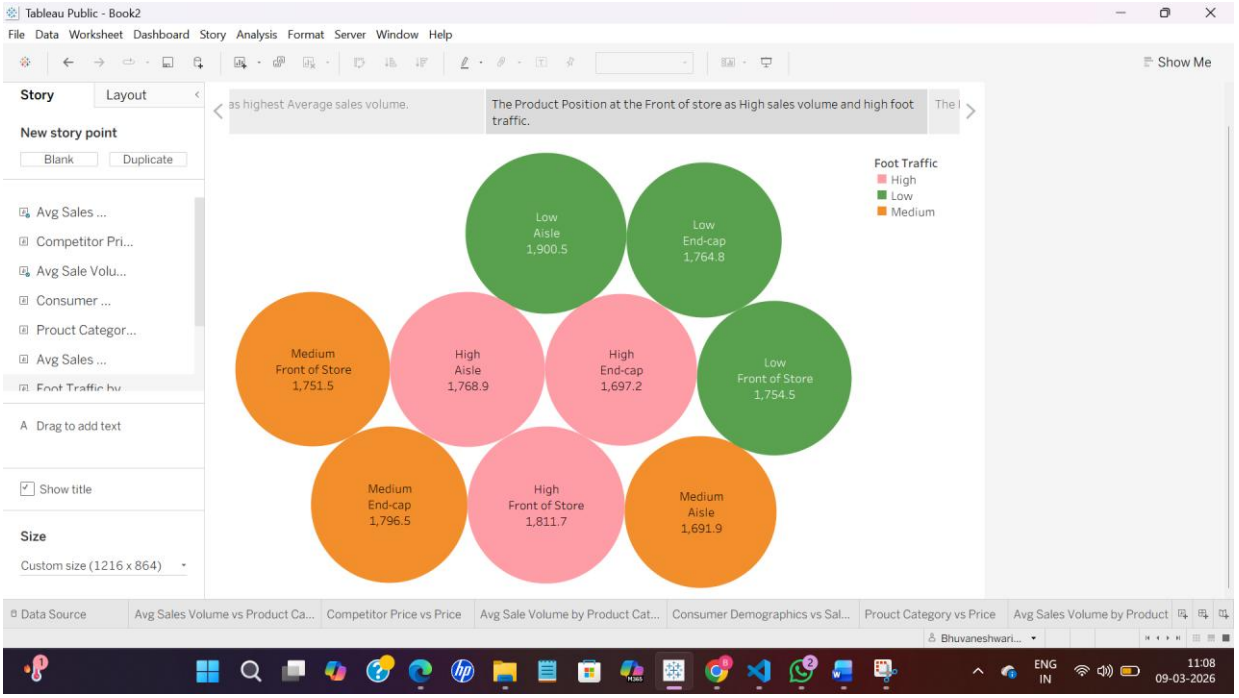


Tableau Public - Book2
File Data Worksheet Dashboard Story Analysis Format Server Window Help
Show Me
Story
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New story point
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Avg Sales ...
Competitor Pri...
Avg Sale Volu...
Consumer ...
Prouct Categor...
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Size
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as highest Average sales volume.
The Product Position at the Front of store as High sales volume and high foot traffic.
Foot Traffic
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Low
Medium
Low
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Low
End-cap
1,764.8
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Front of Store
1,751.5
High
Aisle
1,768.9
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End-cap
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Front of Store
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End-cap
1,796.5
High
Front of Store
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Medium
Aisle
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Data Source
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Competitor Price vs Price
Avg Sale Volume by Product Cat...
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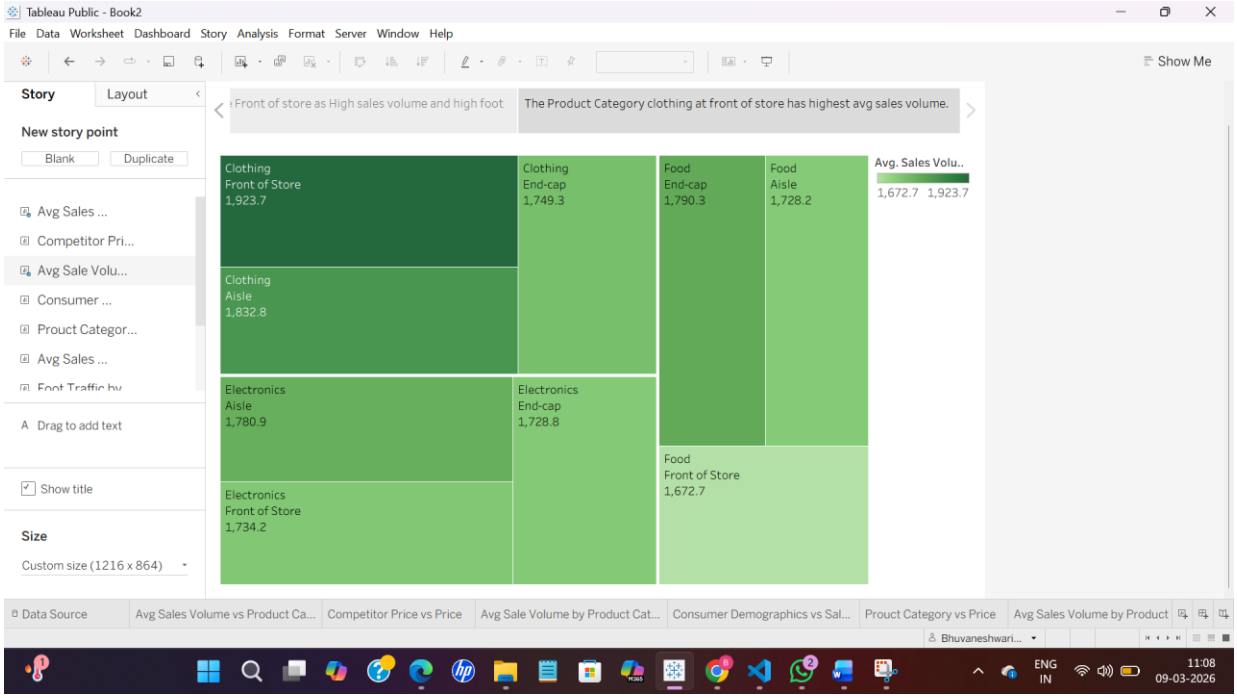
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The Product Category clothing at front of store has highest avg sales volume.
Clothing
Front of Store
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Clothing
End-cap
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Food
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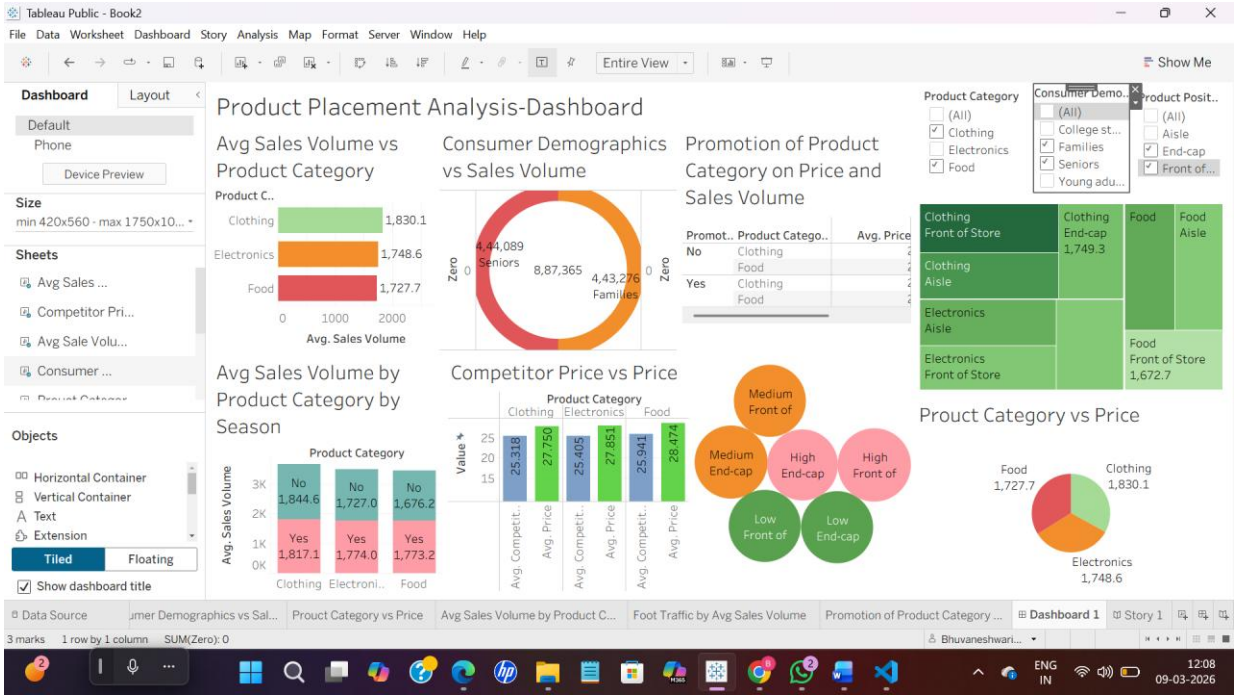
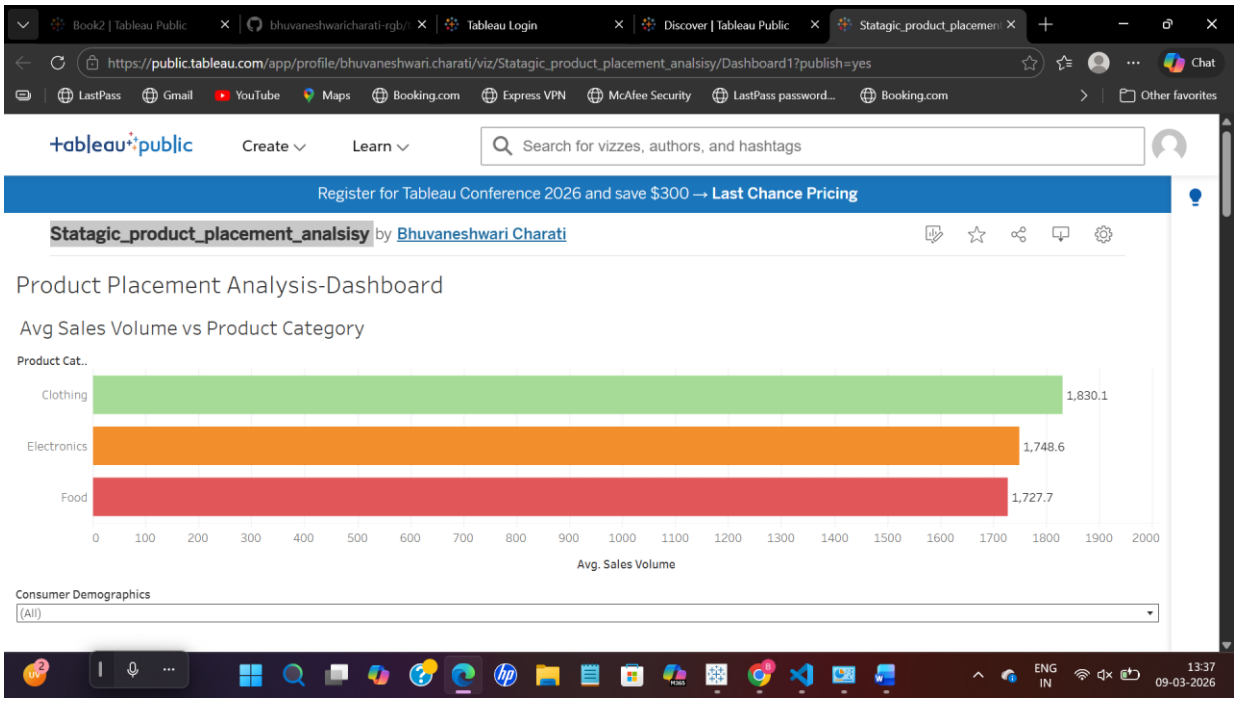
- Filtering

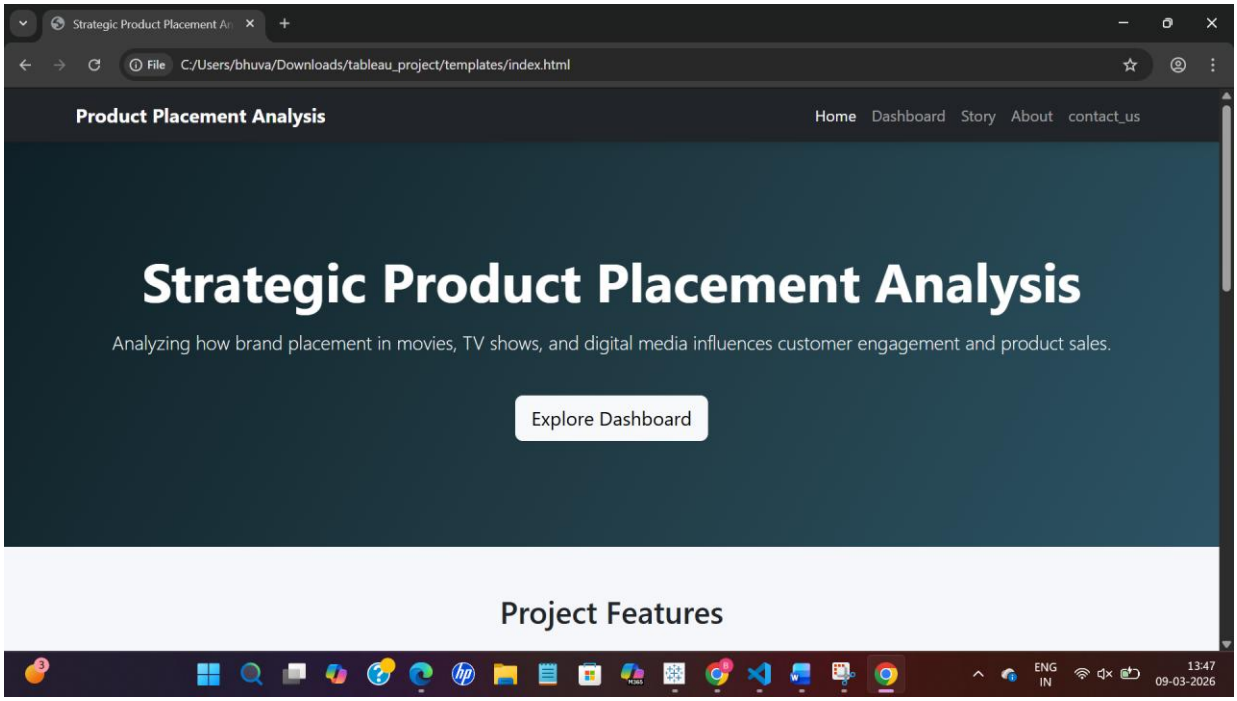
Filters are an indispensable tool in data analysis and visualization, allowing users to refine and focus on specific subsets of data that are relevant to their analysis objectives.

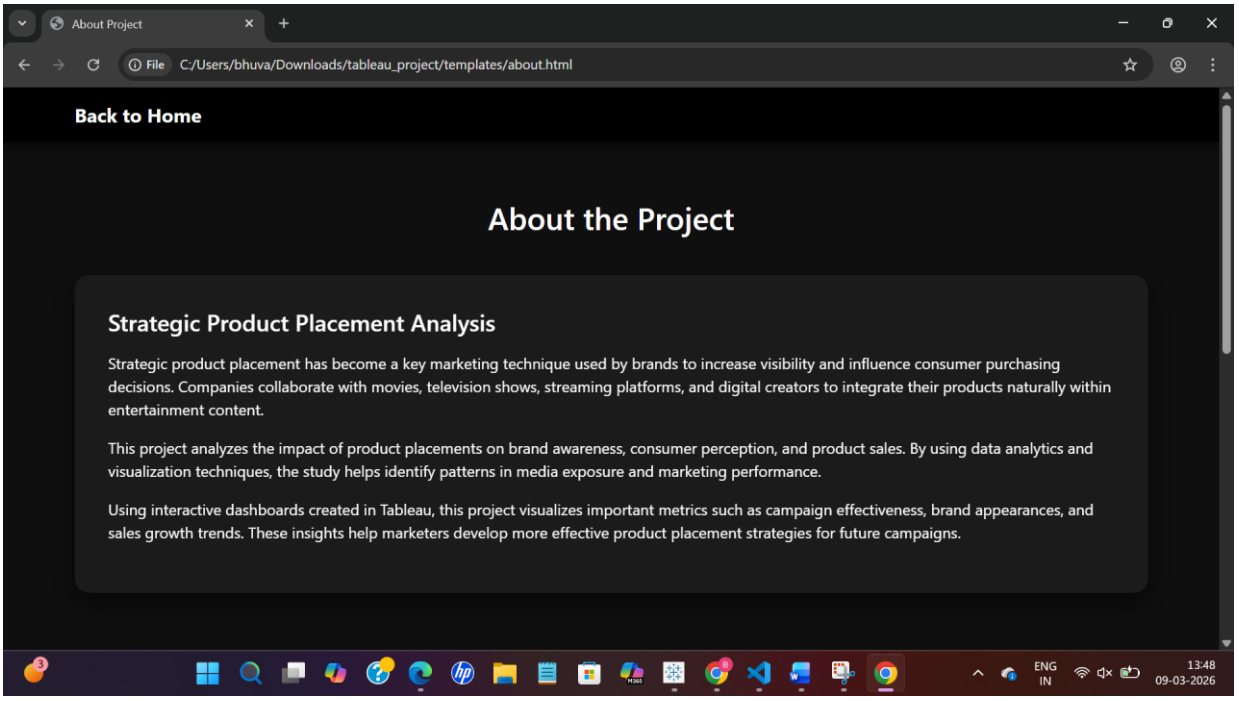


- **Publishing**



- **Dashboard and Story Embed with UI with Flask**




About Project
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About the Project
Strategic Product Placement Analysis
Strategic product placement has become a key marketing technique used by brands to increase visibility and influence consumer purchasing decisions. Companies collaborate with movies, television shows, streaming platforms, and digital creators to integrate their products naturally within entertainment content.
This project analyzes the impact of product placements on brand awareness, consumer perception, and product sales. By using data analytics and visualization techniques, the study helps identify patterns in media exposure and marketing performance.
Using interactive dashboards created in Tableau, this project visualizes important metrics such as campaign effectiveness, brand appearances, and sales growth trends. These insights help marketers develop more effective product placement strategies for future campaigns.
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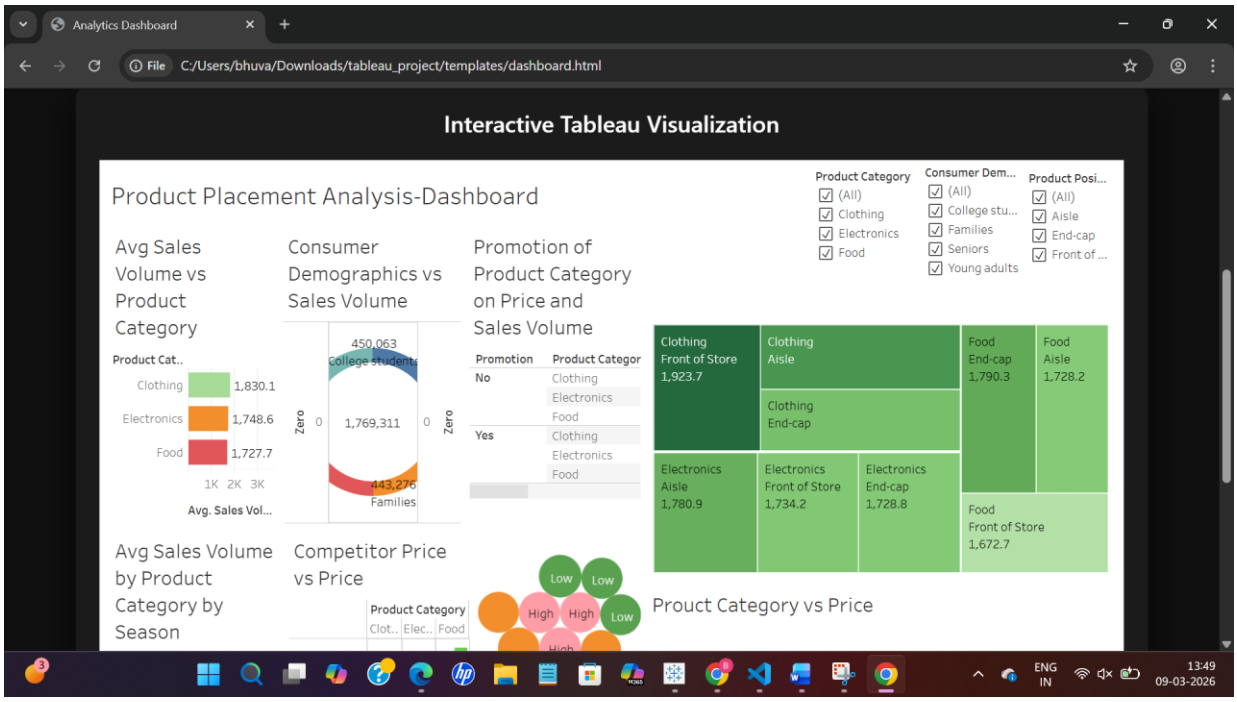
Analytics Dashboard
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Interactive Tableau Visualization
Product Placement Analysis-Dashboard
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Food 1,727.7
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443,276
Families
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Clothing Front of Store 1,923.7
Clothing Aisle
Clothing End-cap
Food End-cap 1,790.3
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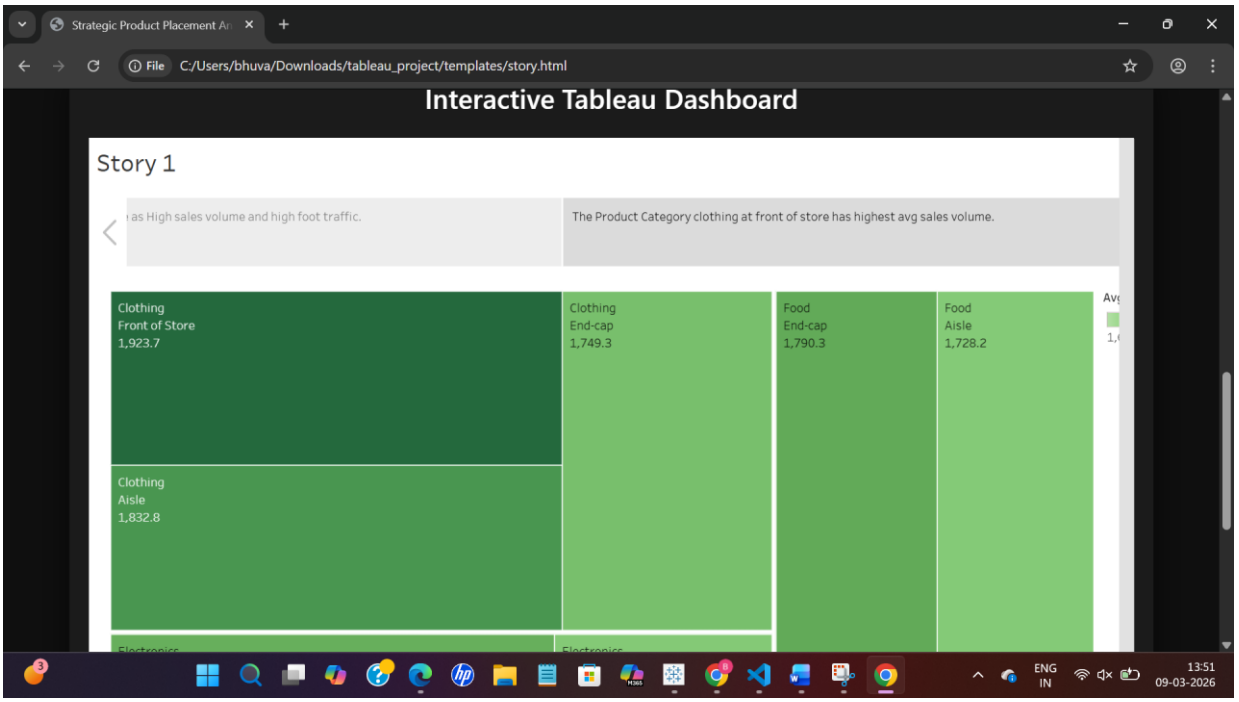
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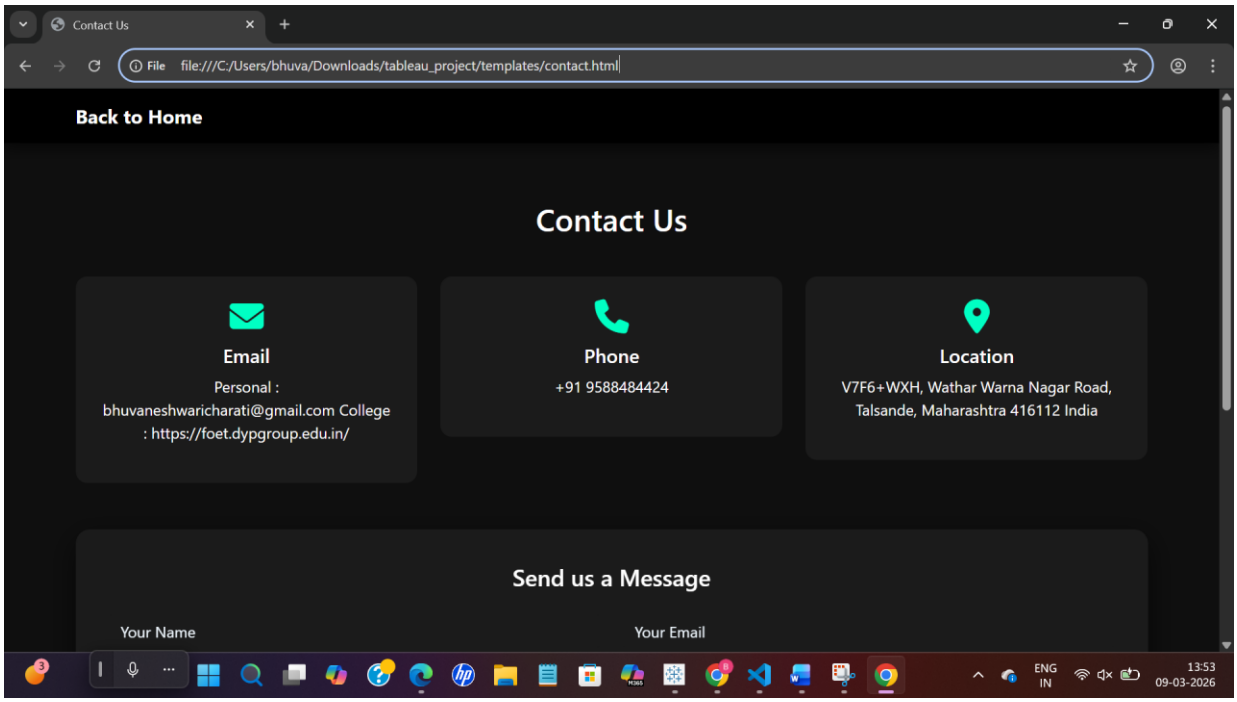
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ADVANTAGES & DISADVANTAGES

Advantages

- Provides clear visualization of complex sales data.
- Helps businesses identify profitable products.
- Supports better strategic decision-making.
- Interactive dashboards make analysis easier.
-

Disadvantages

- Insights depend heavily on the quality of data.
- Limited analysis if dataset is small or incomplete.
- Requires knowledge of Tableau for advanced analysis.

CONCLUSION

The Strategic Product Placement Analysis project serves as an example of how crucial data visualization is to comprehending sales performance. Businesses can quickly spot trends in sales data and assess the success of product placement tactics by utilizing Tableau dashboards.

The project demonstrates how visual analytics can convert unprocessed data into insightful knowledge that enhances data-driven decision-making and boosts overall company performance.

FUTURE SCOPE

Improvements in the future can make this project better. You can connect the system to real-time sales databases to get the most up-to-date information.

You can also use machine learning to guess what sales trends will happen in the future and suggest the best ways to place products.
The project can also be expanded by creating web-based dashboards or connecting it to business intelligence platforms for more advanced analytics.

APPENDIX

- **Kaggle Sales Dataset:**

https://www.kaggle.com/datasets/amitvkulkarni/impact-of-product-positioning-on-sales

- **Tableau Public Dashboard:**

https://public.tableau.com/views/Book2_17728698953120/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

- **GitHub Repository:**

https://github.com/bhuvaneshwaricharati-rgb/tableau_project.git

- **Demo Video:**

https://drive.google.com/file/d/17pyctfRSm6rIkFf7WCkI86jDm66TPpLk/view